



5 It is given that $S_n = \sum_{r=1}^n u_r$, where $u_r = x^{f(r)} - x^{f(r+1)}$ and $x > 0$.

(a) Find S_n in terms of n, x and the function f . [2]

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(b) Given that $f(r) = \ln r$, find the set of values of x for which the infinite series

$$u_1 + u_2 + u_3 + \dots$$

is convergent and give the sum to infinity when this exists. [3]

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